ELM: electroluminescence and light-guide manifold

NAUSICAA design note

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1 Purpose and configuration

The EL manifold (ELM) combines an electroluminescence section with a downstream optical and voltage-grading section. Electrons from the TPC enter holes in an acrylic structure with copper-coated electrode faces. Electroluminescence (EL) is produced inside those holes. Each hole faces a light-guide rod whose entrance is coated with tetraphenyl butadiene (TPB). Converted visible light is transported to a SiPM at the grounded end of the rod. Each rod is surrounded by a continuous xenon-filled annulus. Acrylic fills the space between those gas channels and is treated as opaque to incident VUV light in this design. The gas annulus provides the low-index boundary for total internal reflection (TIR).

The geometry and operating values below are user design inputs. They define a adopted configuration for production and parameter scans. Hole diameter, hole pitch, EL thickness, rod length, SiPM dimensions, pressure, fields, optical properties and sensor response are configurable. One guide and one sensor per EL hole define the initial channel arrangement.

The adopted baseline has 2.5 mm diameter holes and rods on a 4 mm pitch, a 0.5 mm radial xenon gap around each rod, and centred 3×3 mm^2 Hamamatsu S14160-3015PS SiPMs. The TPB conversion efficiency is 0.65. Norland NOA 87 couples the polished rod exit to the SiPM window. Electron collection is set to unity for $\mathcal{E}_{\mathrm{EL}} \geq 2$ kV/cm/bar in this geometry. The optical baseline uses isotropic emission inside the guide, meridional trapping, lossless propagation and unit coupling transmission. These are the agreed calculation assumptions. Figure 1 defines the geometry.

2 Geometry and nominal potentials

Let s increase from the cathode toward the electron readout. The cathode reference is the surface of the ion grid facing the TPC, at potential V_c. For TPC length L and drift-field magnitude E_d, the potential at the end of the drift region is

$$V_d = V_c + E_d L. \tag{1}$$

A transfer gap of length g and nominal field magnitude E_g leads to the EL entry electrode:

$$V_{\mathrm{in}} = V_d + E_g g. \tag{2}$$

The initial values are $V_c = 1.2$ kV, $L = 150$ cm, $E_d = 200$ V/cm, $g = 1$ cm and $E_g = 300$ V/cm. Hence $V_d = 31.2$ kV and $V_{\mathrm{in}} = 31.5$ kV.

Write $\mathcal{E}_{\mathrm{EL}} = E_{\mathrm{EL}}/P$ for the reduced EL field, P for pressure and h for the EL thickness. Then

$$E_{\mathrm{EL}} = P\mathcal{E}_{\mathrm{EL}}, \qquad \Delta V_{\mathrm{EL}} = P\mathcal{E}_{\mathrm{EL}} h, \qquad V_{\mathrm{out}} = V_{\mathrm{in}} + P\mathcal{E}_{\mathrm{EL}} h. \tag{3}$$

Using P in bar, $\mathcal{E}_{\mathrm{EL}}$ in kV/cm/bar and h in cm gives potentials in kV. At $P = 15$ bar, $h = 5$ mm and $\mathcal{E}_{\mathrm{EL}} = 2$ kV/cm/bar, the EL field is 30 kV/cm, the EL voltage difference is 15 kV, and the exit electrode is at $V_{\mathrm{out}} = 46.5$ kV. The intended scan begins at 2 and extends to 3 kV/cm/bar.

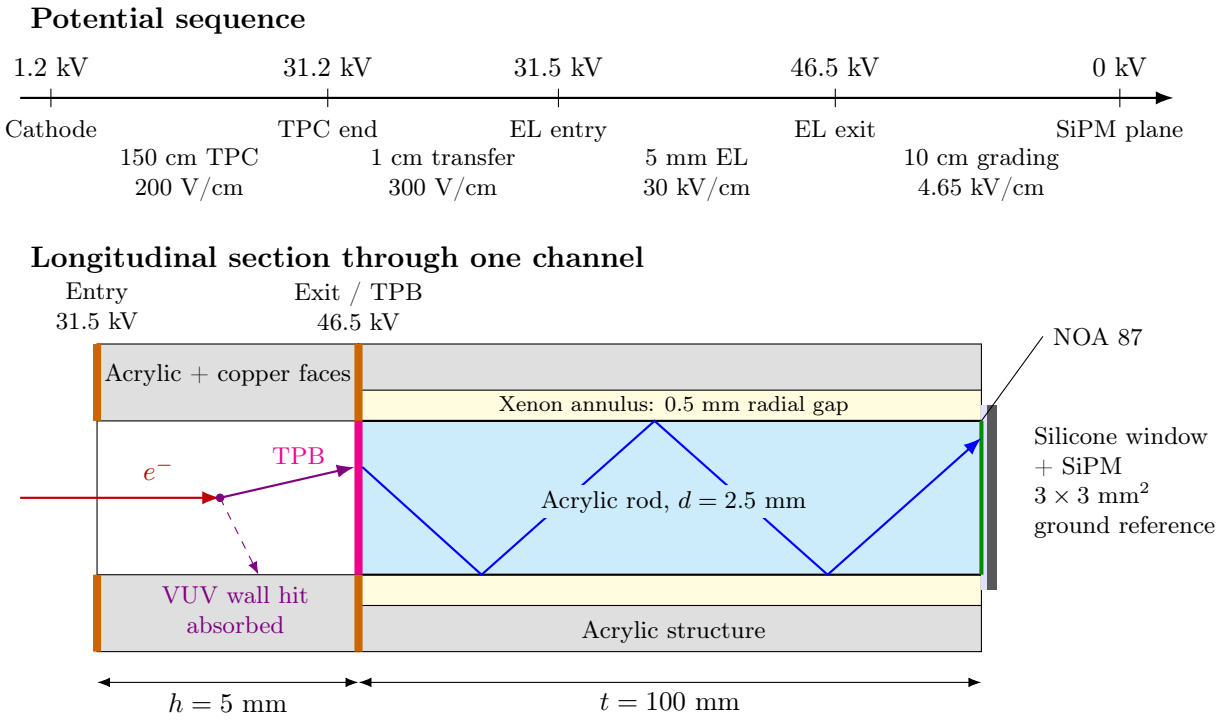


Figure 1: ELM baseline at $E/P = 2$ kV/cm/bar. Top: voltage sequence from the ion-grid cathode to the SiPM reference plane. Centre: longitudinal section through one EL hole and its guide. Rod diameter and radial gas gap are shown in proportion; longitudinal lengths and sensor thickness are schematic. Bottom: neighbouring guide channels at 4 mm pitch. The gas annulus is 0.5 mm thick radially, leaving a 0.5 mm minimum acrylic web. The blue ray illustrates TIR; actual directions are sampled from the adopted angular model. The exit-electrode aperture and grading electrodes require a detailed electrostatic design; their nominal potential boundary is shown.

The downstream rod section has length t. Its entrance is associated with the EL exit potential, and its far end with the readout-plane reference $V_s = 0$. Its required average potential gradient magnitude is

$$G_{\mathrm{M}} = \frac{V_{\mathrm{out}} - V_s}{t}. \tag{4}$$

At $t = 10$ cm, $G_{\mathrm{M}} = 4.65$ kV/cm. At 15 bar the corresponding reduced-field reference is 0.31 kV/cm/bar. The design objective is to grade the voltage without producing additional EL in gas exposed to the downstream fields. Equation (4) specifies an average gradient; the grading electrodes and dielectric interfaces determine the local field. The reduced-field reference applies to gas regions, while fields in the solid rods and filler are described by their dielectric boundary conditions.

The nominal potential rises along the drift and EL regions and falls along the rod section. Since $\mathbf{E} = -\nabla V$, the electric-field direction reverses downstream of the EL exit. Electron collection is assigned to the EL exit electrode; photons continue through the optical section. The exit electrode's optical transmission is a configurable part of the light budget. SiPM bias is referenced locally to the grounded readout plane.

Region	Length	Entrance potential	Exit potential
TPC drift	150 cm	1.2 kV	31.2 kV
Transfer gap	1 cm	31.2 kV	31.5 kV
EL section, $\mathcal{E}_{\mathrm{EL}} = 2$	0.5 cm	31.5 kV	46.5 kV
Guide and grading section	10 cm	46.5 kV	0 kV

3 Linking guide length to the EL operating point

Two independent inputs suffice to specify any of the combinations in equation (4). The card selects a length policy:

Fixed length. Specify t and derive G_{M} at the chosen $\mathcal{E}_{\mathrm{EL}}$.

Fixed gradient. Specify a target G_{M} and derive

$$t(\mathcal{E}_{\mathrm{EL}}) = \frac{V_c + E_d L + E_g g + P\mathcal{E}_{\mathrm{EL}} h - V_s}{G_{\mathrm{M}}}. \tag{5}$$

This is the initial interpretation of keeping the voltage-grading field the same while changing $\mathcal{E}_{\mathrm{EL}}$.

Fixed reduced gradient. If pressure is scanned, an alternative policy specifies $\mathcal{G} = G_{\mathrm{M}}/P$ and uses $t = (V_{\mathrm{out}} - V_s)/(P\mathcal{G})$.

At fixed pressure, the latter two policies coincide after matching their reference values. With the starting geometry and $G_{\mathrm{M}} = 4.65$ kV/cm, equations (3) and (5) give:

$\mathcal{E}_{\mathrm{EL}}$ (kV/cm/bar)	ΔV_{EL} (kV)	V_{out} (kV)	t (cm)
2.0	15.00	46.50	10.000
2.5	18.75	50.25	10.806
3.0	22.50	54.00	11.613

These are nominal voltage-budget calculations. A longer guide also changes optical propagation and attenuation; the derived length belongs to both the electrical geometry and the optical response.

4 Holes, gas annuli and sensors

The hole and guide diameter is $d = 2.5$ mm, their common pitch is $p = 4$ mm, and the SiPM active area is 3×3 mm^2. A centred circular rod therefore fits inside the sensor footprint. Each guide has a continuous xenon annulus of radial thickness $\delta = 0.5$ mm. Its outer diameter and the minimum acrylic web between nearest neighbours are

$$D_{\text{channel}} = d + 2\delta = 3.5 \text{ mm}, \qquad w = p - D_{\text{channel}} = 0.5 \text{ mm}. \tag{6}$$

The polished rod–gas interface provides TIR. Gap thickness controls the mechanical geometry; its optical role is to preserve that interface continuously along the rod. Optical glue is confined to the downstream end face. The sensor footprint here denotes its active area. The adopted design uses a compact SiPM matrix with 3 mm × 3 mm active areas on a 4 mm pitch (user design input). The S14160-3015PS supplies the response parameters; its individual package has a 4.35 mm dimension [1]. The matrix packaging is to be designed for the adopted pitch. The optical calculation uses the centred active areas shown in the drawing.

The lattice remains configurable. The aperture ratio, defined as hole area divided by area per cell, is

$$f_{\text{open,sq}} = \frac{\pi d^2}{4p^2} = 30.68\%, \qquad f_{\text{open,hex}} = \frac{\pi d^2}{2\sqrt{3}p^2} = 35.43\%. \tag{7}$$

Figure 3 of the AXEL prototype study [2] motivates small pitch and aperture ratios above the region of poor electron collection (approximately 0.2 in their scan). That study uses $E/P = 3$ kV/cm/bar, 5 mm EL depth, and its own gas and optical conditions. For the present baseline, the adopted electron collection efficiency is

$$\epsilon_{\text{coll}} = 1 \quad \text{for} \quad \mathcal{E}_{\text{EL}} \geq 2 \text{ kV/cm/bar}. \tag{8}$$

This is a design approximation for the agreed geometry. At the starting operating point, the nominal hole-to-transfer-field ratio is $E_{\text{EL}}/E_g = 100$. A later field calculation can replace ϵ_{coll} with a position-dependent efficiency including wall interception.

5 EL production and direct VUV acceptance

5.1 Mean EL yield

The existing implementation, `el_photon_yield` in `src/event_analysis/grid_readout.jl`, evaluates

$$Y_g = a_g \max(\mathcal{E}_{\text{EL}} - x_{0,g}, 0)\, P h_{\text{cm}} \quad \text{photons per entering electron}. \tag{9}$$

Here P is in bar, $\mathcal{E}_{\text{EL}}$ and x_0 in kV/cm/bar, and a_g in photons/electron/kV. The fits in Table 1 of Ref. [3], as stored in `config/electron_transport.toml`, are

Gas	a_g	$x_{0,g}$ (kV/cm/bar)
Pure Xe	137	0.73
Xe–15% He	139	0.80

Their use at the selected pressure is the operating-pressure scaling already adopted by the simulation. Electron counts entering this calculation are those surviving drift attachment. Collection into the hole then uses ϵ_{coll} above.

5.2 Solid angle and absorbing walls

Take isotropic VUV emission uniformly along the hole axis. Let $r = d/2$ and let u be the axial distance from an emission point to the exit disk. The accepted half-angle is $\alpha(u) = \arctan(r/u)$, so the fraction of all emitted photons reaching the exit is

$$A(u) = \frac{2\pi[1 - \cos\alpha(u)]}{4\pi} = \frac{1}{2}\left(1 - \frac{u}{\sqrt{u^2 + r^2}}\right). \tag{10}$$

The straight segment joining an axial emission point to this disk remains inside the cylindrical hole. Photons heading backward or intersecting a wall are absorbed. Averaging along the hole gives

$$A_{\text{exit}}(r, h) = \frac{1}{h}\int_0^h A(u)\, du \tag{11}$$

$$= \frac{h + r - \sqrt{h^2 + r^2}}{2h} = \frac{r}{h + r + \sqrt{h^2 + r^2}}. \tag{12}$$

The last form is used by `axial_light_acceptance`. The baseline assumes unit transmission of the exit-electrode opening; an additional electrode transparency $T_{\text{electrode}}$ can multiply Eq. (12). For nonuniform axial emission density $q(u)$, replace the uniform average by $\int q(u)A(u)du / \int q(u)du$. The worked example uses the uniform-field, uniform-yield prescription.

6 TPB conversion and guide trapping

6.1 Conversion at the entrance

All VUV photons reaching the coated guide face are taken to be absorbed. The adopted conversion efficiency is $\eta_{\text{TPB}} = 0.65$ visible photons per absorbed VUV photon. This is the user-selected material input; measurements of TPB intrinsic efficiency and its wavelength dependence are reported by Benson et al. [4]. Re-emission is isotropic, giving a forward fraction $f_+ = 1/2$:

$$\langle N_{\text{blue},+}\rangle = N_{\text{VUV,exit}}\, \eta_{\text{TPB}} f_+. \tag{13}$$

The forward factor is applied once. The VUV incidence angle is integrated out by absorption and re-emission under this model.

6.2 Analytical trapping cone

Let n_g denote the rod refractive index and n_c the surrounding xenon index. Use $n_g = 1.50$ and $n_c = 1.00$ as rounded design inputs. Cast acrylic has $n = 1.502$ at 436 nm in the manufacturer's data [5]; the adopted gas index approximates its value by unity. For a meridional ray at angle θ to the guide axis, incidence on the cylindrical wall is $i = \pi/2 - \theta$. Snell's law gives

$$i > \arcsin(n_c/n_g), \qquad \theta < \theta_{\max} = \arccos(n_c/n_g). \tag{14}$$

For an isotropic angular distribution inside the rod, the fraction of the forward hemisphere in this cone is

$$f_{\text{trap}|+} = \frac{\int_0^{2\pi} d\phi \int_0^{\theta_{\max}} \sin\theta\, d\theta}{2\pi} = 1 - \frac{n_c}{n_g} = 0.3333. \tag{15}$$

Here $\theta_{\max} = 48.19$ degrees. Relative to all re-emitted blue photons, the guided fraction is $f_+ f_{\text{trap}|+}$. This is the meridional trapping approximation discussed in Ref. [6]. It is exact for the directional acceptance of an on-axis source in an ideal cylindrical rod. Applying this cone to the extended TPB-coated face is our baseline approximation; off-axis skew rays can add guided light. The interface treatment assumes that the forward blue light enters the core with the stated isotropic distribution. A resolved TPB–core model would propagate the angular distribution through that interface.

6.3 Propagation over the guide length

For ideal polished walls, each TIR reflection preserves the photon. The baseline sets $T_{\mathrm{prop}} = 1$. A finite bulk attenuation length $\Lambda(\lambda)$ can be introduced explicitly. A ray travels $\ell = t/\cos\theta$, so its survival is

$$T_{\mathrm{bulk}}(\theta, \lambda) = \exp[-t/(\Lambda(\lambda)\cos\theta)]. \tag{16}$$

For the adopted angular cone and normalized blue emission spectrum $S(\lambda)$,

$$\overline{T}_{\mathrm{prop}} = \frac{1}{f_{\mathrm{trap}|+}} \int d\lambda\, S(\lambda) \int_0^{\theta_{\max}} \sin\theta\, \exp[-t/(\Lambda(\lambda)\cos\theta)]\, d\theta. \tag{17}$$

This expression uses wavelength-independent design indices; dispersion can be included by evaluating the trapping angle at each wavelength. With $\Lambda \to \infty$, it returns unity. Changing t changes attenuation and optical flight time even when the trapping cone is unchanged.

7 NOA 87 coupling and packaged SiPM PDE

Norland NOA 87 is the selected coupling adhesive between the rod exit and the S14160-3015PS entrance window. This device has a silicone-resin window with nominal refractive index 1.57 [1]. The manufacturer quotes a cured refractive index of 1.52, excellent glass adhesion and good-to-excellent plastic adhesion [7]. The liquid index is 1.49; calculations of the completed joint use the cured material. Its wavelength dispersion, absorption and bond thickness remain configurable optical inputs.

For an incoherent adhesive layer at normal incidence, interface reflectance is

$$R_{ij} = \left(\frac{n_i - n_j}{n_i + n_j}\right)^2, \qquad T_{\mathrm{interfaces}} \simeq (1 - R_{ga})(1 - R_{as}), \tag{18}$$

where a denotes adhesive and s the silicone-resin window. The optimum intermediate index is approximately $n_a = \sqrt{n_g n_s}$. With the adopted $n_g = 1.50$ and nominal $n_s = 1.57$, this gives $n_a = 1.535$. NOA 87 provides a close optical match; its nominal cured index of 1.52 is the material choice for this design. The baseline sets the added external coupling transmission to unity.

The SiPM's packaged-device PDE is the probability of detecting a photon incident on its entrance aperture, including its window response [8]. The separate window-transmission factor is set to unity in our bookkeeping. The photon-to-primary-avalanche conversion is

$$N_{\mathrm{PE}} \mid N_{\gamma,\mathrm{SiPM}} \sim \mathrm{Binomial}(N_{\gamma,\mathrm{SiPM}}, \overline{\mathrm{PDE}}), \qquad \overline{\mathrm{PDE}} = \int S_{\mathrm{arr}}(\lambda)\mathrm{PDE}_{\mathrm{package}}(\lambda, V_{\mathrm{ov}})\, d\lambda. \tag{19}$$

The arriving spectrum S_{arr} is normalized and includes any spectral filtering in the guide. A later explicit calculation of the coupled window boundary must replace the corresponding air–window response in the PDE calibration consistently. Recovery, correlated avalanches, saturation and electronics follow this photon-detection step.

8 Complete optical chain and fluctuations

With one rod and sensor per hole, the mean probability of producing a blue photon at the SiPM entrance from an emitted EL photon is

$$q_\gamma = A_{\mathrm{exit}}\, T_{\mathrm{electrode}}\, \eta_{\mathrm{TPB}}\, f_+\, f_{\mathrm{trap}|+}\, \overline{T}_{\mathrm{prop}}\, T_{\mathrm{coupling}}\, f_{\mathrm{coverage}}. \tag{20}$$

The centred 2.5 mm circular exit lies fully inside the 3×3 mm^2 active footprint, so $f_{\mathrm{coverage}} = 1$ for the thin, ideally coupled joint. The baseline models absorption/conversion as a Bernoulli probability

0.65 and the other stages as independent photon survival probabilities. Consequently, conditional on the emitted EL count,

$$N_{\gamma,\mathrm{SiPM}} \mid N_{\mathrm{EL}} \sim \mathrm{Binomial}(N_{\mathrm{EL}}, q_\gamma). \tag{21}$$

This thinning follows the existing primary-charge and EL fluctuation stages. For N_e electrons entering a hole, use

$$\langle N_{\mathrm{EL}} \mid N_e \rangle = N_e Y_g, \qquad \mathrm{Var}(N_{\mathrm{EL}} \mid N_e) = N_e J Y_g. \tag{22}$$

J is defined in Ref. [9]; the agreed approximate reading is $J = 0.030$ at $E/P = 2$, shared by the two gases as a model input. The resulting conditional moments at the SiPM entrance are

$$\langle N_{\gamma,\mathrm{SiPM}} \mid N_e \rangle = N_e Y_g q_\gamma, \tag{23}$$

$$\mathrm{Var}(N_{\gamma,\mathrm{SiPM}} \mid N_e) = N_e Y_g \left[q_\gamma(1 - q_\gamma) + J q_\gamma^2 \right]. \tag{24}$$

Primary Fano fluctuations and drift-attachment fluctuations are already carried by N_e. These formulas preserve the separate contributions of EL fluctuations and optical thinning.

The saved optical output is blue photons per SiPM and arrival-time bucket. The initial width is 0.25 μs. The baseline assigns photons to the parent electron arrival bucket, retaining the existing timing convention. The subsequent SiPM response applies the selected PDE and electronics model.

9 Worked example: $E/P = 2$ kV/cm/bar

Take $P = 15$ bar, $h = 5$ mm, $d = 2.5$ mm, $p = 4$ mm and $t = 10$ cm. The cathode, TPC and transfer-gap inputs give

$$V_d = 1.2 + 0.200 \times 150 = 31.2 \text{ kV}, \tag{25}$$

$$V_{\mathrm{in}} = V_d + 0.300 \times 1 = 31.5 \text{ kV}, \tag{26}$$

$$E_{\mathrm{EL}} = 15 \times 2 = 30.0 \text{ kV/cm}, \tag{27}$$

$$V_{\mathrm{out}} = V_{\mathrm{in}} + 15 \times 2 \times 0.5 = 46.5 \text{ kV}, \tag{28}$$

$$G_{\mathrm{M}} = V_{\mathrm{out}}/10 = 4.65 \text{ kV/cm}. \tag{29}$$

The corresponding reduced grading reference is 0.31 kV/cm/bar.

For $r = 1.25$ mm, Eq. (12) gives $A_{\mathrm{exit}} = 0.109612$. With $\eta_{\mathrm{TPB}} = 0.65$, $f_+ = 0.5$, $f_{\mathrm{trap}|+} = 0.3333$ and unit electrode, propagation, coupling and coverage factors,

$$q_\gamma = 0.109612 \times 0.65 \times 0.5 \times \left(1 - \frac{1}{1.5}\right) = 0.011875 = 1.187\%. \tag{30}$$

The mean light budget is:

Stage	Photons per 1000 emitted EL photons
At hole exit / TPB	109.61
Blue re-emission, both hemispheres	71.25
Blue re-emission, forward hemisphere	35.62
Guided blue light at SiPM entrance	11.87

Using Eq. (9) for the two gas models:

Mean photons per entering electron	Pure Xe	Xe–15% He
Emitted EL photons	1,304.92	1,251.00
VUV photons reaching TPB	143.04	137.12
Blue photons at SiPM entrance	15.50	14.86
Primary PE, PDE = 0.32	4.96	4.75

The packaged PDE and operating point are stored in the response card. The 1.187% optical estimate is the result of the stated axial-emission and meridional-capture model with ideal propagation and coupling.

10 Sensor, timing and electronics baseline

The selected S14160-3015PS has a 3×3 mm^2 active area and 15 μm microcell pitch, with 39,984 cells. Its nominal gain is 3.6×10^5, peak packaged PDE is 0.32 at 460 nm, and primary dark rate is 700 kcps at 25°C and +4 V overvoltage [1]. The simulation uses this peak PDE as a provisional average over the arriving TPB spectrum.

The circular 2.5 mm guide illuminates a centred disk on the sensor. Uniform illumination of that footprint gives

$$M_{\mathrm{lit}} = \mathrm{round}\left(M \frac{\pi(d/2)^2}{(3\ \mathrm{mm})^2}\right) = 21{,}808. \tag{31}$$

The remaining 18,176 cells contribute dark avalanches. The full dark rate is split between the populations in proportion to their cell counts. The statistical recovery simulation uses uniform photon arrival times inside each 250 ns bucket. Its baseline exponential recovery constant is 12.5 ns, with 25 ns as a sensitivity case. These are provisional model inputs: the 12.5 ns proxy originated from an output-pulse decay measurement for the smaller S14160-1315PS [10]. The current response excludes crosstalk, afterpulsing and gain spread by explicit card settings.

The ASIC input is the bucket-average current. For fully recovered avalanche charge $q_{\mathrm{PE}} = eG$, the upper current limit maps to

$$N_{\mathrm{PE,limit}} = \frac{I_{\max}\Delta t}{eG} = 4{,}334.4\ \mathrm{PE}, \qquad I_{\max} = 1\ \mathrm{mA}. \tag{32}$$

Independent Gaussian electronics noise has RMS 3.5417 PE per bucket, with signed output retained and no lower clipping. This is the same input charge noise as 0.5 PE at gain 2.55×10^6, expressed at the selected gain. The limit and noise are design inputs.

The matched ion–electron production samples one ion reference drift distance uniformly from 5 to 145 cm and sets the electron distance to its complement within the 150 cm chamber (user design input). Both arms read the same saved placement and primary-pair realization. The ion grid has 4 mm pitch and 40 ms buckets; the electron grid has 4 mm pitch and 250 ns buckets. Electron speed is 1 mm/μs and lifetime is 15 ms. Photons retain the parent electron arrival bucket. Event-energy comparisons correct known attachment survival before evaluating the remaining fluctuations.

11 Pilot comparison and adopted geometry

The user selected the 4 mm pitch, 2.5 mm diameter geometry after matched bb0ν pilots in Xe–15% He. Each configuration uses 100 events, $E/P = 2$ kV/cm/bar, the same 10–150 cm drift interval, optical assumptions and electronics settings. The earlier 5 mm pitch, 3 mm diameter pilot remains available with its original cards and outputs. The table is generated from their saved HDF5 reductions.

Pitch (mm)	d (mm)	τ_r (ns)	SiPM loss (%)	ASIC loss (%)	Retained events	Full / retained RMS (%)
5	3	12.5	0.0790	0.2184	89/100	0.9396 / 0.2258
5	3	25	0.1576	0.2098	90/100	0.9521 / 0.2222
4	2.5	12.5	0.0673	0.0165	97/100	0.3360 / 0.2928
4	2.5	25	0.1343	0.0136	97/100	0.3349 / 0.2777

SiPM loss is summed optical charge missing through incomplete recovery, divided by summed primary PE. ASIC loss is clipped charge divided by charge entering the ASIC. Retained events have no clipped acquired bucket. Full and retained RMS values are $100\,\sigma/\mu$, evaluated over all events and the retained subset respectively; they are not FWHM. Each event total is divided by its known attachment survival and deposited MC energy. Dark expectation is subtracted, and signed electronics noise is summed over the complete acquired window.

At the baseline recovery time, the geometry change reduces ASIC charge loss from 0.2184% to 0.0165%, and raises the retained count from 89 to 97. The full-sample RMS changes from 0.9396% to 0.3360%. The smaller aperture and finer pitch reduce light and charge per channel: the mean of event-mean primary PE pulses changes from 121.65 to 76.23, while the mean of event maxima changes from 1,994.41 to 1,249.83. These pulse statistics use the same positive photon-input SiPM–bucket cells throughout the response.

The tradeoff is visible in the RMS with clipping disabled, which changes from 0.2347% to 0.2942% at 12.5 ns. More channels and fewer detected photons increase the relative contribution of electronics noise and photon statistics. The adopted geometry prioritizes reduced clipping and finer spatial sampling. These small pilots establish the comparison; their numerical results are simulated estimates for the stated model.

12 Configuration parameters

The matched detector configuration is `config/specs/detector/configurations/elm_matched_full.toml`, with shared placement in `config/specs/realisations/elm_matched_full.toml` and electron settings in `config/specs/detector/electrons/elm_4mm_250ns_ep2_shared.toml`; the baseline sensor card is `config/specs/response/sipm/elm_pitch4mm_3015ps_tau12p5ns.toml`. The electronics card is `config/specs/response/sipm_electronics/readout_s14160_1315ps.toml`; its charge-noise and current-limit settings also apply to the 3015PS at the same gain. The table summarizes these inputs together with the design-note parameters used to derive the electrode potentials.

Parameter	Adopted value or role
`pressure_bar`, `helium_fraction`	15; 0 or 0.15 for the two gas models
`chamber_length_cm`, `drift_field_V_cm`	150; 200
`cathode_potential_V`	1200, ion-grid TPC-facing surface
`entrance_gap_mm`, `entrance_field_V_cm`	10; 300
`hole_depth_mm`	5
`reduced_field_kV_cm_bar`	2.0 initially; scan to 3.0
`hole_pitch_mm`	4; square grid in the readout implementation
`hole_diameter_mm`	2.5
`guide_diameter_policy`	Equal to hole diameter initially
`gas_gap_radial_mm`	0.5; xenon surrounds the rod
`guide_refractive_index`, `gap_refractive_index`	1.50; 1.00, blue-light approximations
`tpb_conversion_efficiency`	0.65; full VUV absorption
`electron_collection_efficiency`	1 for the baseline at $E/P \geq 2$
`guide_attenuation_length_cm`	Infinite in the baseline
`coupling_adhesive`	Norland NOA 87; nominal cured $n = 1.52$
`coupling_transmission`, `electrode_transmission`	1; 1 in the baseline
`pde_convention`	Packaged-device PDE, including window
`guide_length_policy`	Fixed length, fixed gradient, or fixed reduced gradient
`guide_length_mm`	100 when fixed-length mode is selected
`grading_field_kV_cm`	4.65 when fixed-gradient mode is selected
`grading_reduced_field_kV_cm_bar`	0.31 when fixed-reduced-gradient mode is selected
`sensor_plane_potential_V`	0
`sipm_shape`, `sipm_side_mm`	Square; 3
`bucket_width_us`	0.25 initially
`min_ion_drift_cm`, `max_ion_drift_cm`	5; 145 in the common realization; electron drift is the complement
`lifetime_ms`, `drift_speed_mm_us`	15; 1
`microcell_count`, `illuminated_microcell_count`	39,984; 21,808
`pde`, `gain`	0.32; 3.6×10^5

Parameter	Adopted value or role
`recovery_s`	12.5×10^{-9}; sensitivity 25×10^{-9}
`current_limit_A`, `noise_rms_pe`	0.001; 3.5417
Guide material, cladding, optical surfaces	Refractive indices, attenuation, coupling and alignment
TPB coating	Absorption, conversion efficiency, emission spectrum and timing
Exit electrode and grading electrodes	Optical transmission and electrostatic boundary geometry
SiPM and electronics	Device parameters, bias, response and noise cards

Resolve the selected policy into a single geometry and potential profile before simulation. Store both the requested card and the derived values in output provenance. Potentials are derived from field/length inputs in this convention; guide length is derived when a grading-gradient policy is selected. Changing $\mathcal{E}_{\mathrm{EL}}$ then changes both EL production and, when linked, guide length. Those dependencies belong to the optical-output identity. Sensor/electronics changes reuse photon histories when the collecting geometry is unchanged.

13 Reproducing the example

The Julia calculation in `scripts/documentation/calculate_elm_example.jl` uses the existing EL-yield and axial-acceptance functions and saves the inputs, results, source hashes and quadrature checks to `results/elm/elm_example.h5`. It independently integrates the exit solid angle and trapping cone to check the analytical expressions. Python formats the saved values into `latex_generated/detector/elm/elm_values.tex`; numerical tables above read those macros. The worked example is deterministic. The calculator also imports scalar reductions from `results/elm_sipm_pitch4mm_pilot/elm_sipm_pitch4mm_pilot.h5`, including its matched 5 mm reference from `results/elm_sipm_pilot/elm_sipm_pilot.h5`. Source hashes and the current detector, sensor and electronics cards are stored with the example. Python formats the pilot table without recomputing its statistics.

From the repository root, with the project environments active:

```
julia -project=.  scripts/documentation/calculate_elm_example.jl
python scripts/documentation/present_elm_example.py
latexmk -pdf -outdir=work/elm_build docs/elm.tex
cp work/elm_build/elm.pdf latex_generated/detector/elm/elm.pdf
```

Presentation regeneration starts with the Python command using the existing HDF5 result. The retained TeX macros also allow the note to compile directly. The vector drawing is editable in `docs/figures/elm_setup.tex`.

References

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